Chapter 1

# Study with structure

Reading source

**Original template example / 模板原创示例**

Heat equation: one model, several ways to understand it.

**Learning objectives**

- ▷ Identify the equation, initial data and boundary values.
- ▷ Relate a sine mode to an energy estimate.
- ▷ Reconstruct a proof before opening the solution.

Reading progress 3/8

## 1.1 References that know their type

The **energy method**（能量方法）turns a differential equation into an estimate for an integral quantity.

**Definition 1.1** (Energy)
For a sufficiently smooth function $u$, set $E(t) = \frac{1}{2}\int_0^L |u(x,t)|^2 \, dx$.

**Theorem 1.2** (Dissipation)
*For a classical heat solution with zero endpoint values,* $E'(t) = -\int_0^L |u_x(x,t)|^2 \, dx \le 0$.

**Lemma 1.3** (A sine mode)
*If* $k = \pi/L$ *and* $u(x,t) = a(t)\sin(kx)$, *then* $u_t = u_{xx}$ *is equivalent to* $a'(t) = -k^2 a(t)$.

Definition 1.1 introduces the quantity used in theorem 1.2. The reduction in lemma 1.3 suggests an explicit example. Labels keep their types while sharing one chapter-based numerical sequence.

## 1.2 Practice before looking at the answer

**Exercise 1.4** (The first mode)

Solve $u_t = u_{xx}$ on $0 < x < L$ with zero endpoint values and $u(x,0) = \sin(\pi x/L)$. Compute the decay rate of $E(t)$.

**Hint / exercise 1.4**

Start with the separated form $u(x,t) = a(t)\sin(\pi x/L)$. Apply lemma 1.3 and use $a(0) = 1$.

**Solution / exercise 1.4**

The time factor decays exponentially. Integrating $a'(t) = -k^2 a(t)$ with $a(0) = 1$ gives

$$u(x,t) = e^{-k^2 t}\sin(kx), \qquad k = \frac{\pi}{L}. \tag{1.1}$$

The energy is $E(t) = \frac{L}{4}e^{-2k^2 t}$.

Python / Decay rate

```
from math import pi
decay_rate = (pi / L)**2
energy_rate = 2 * decay_rate
```

**Common pitfall**

The amplitude and energy decay at different rates. Squaring the amplitude doubles the exponent. 注意区分振幅与能量的衰减率。

## 1.3 A deliberate review

**Key takeaways**

- A boundary condition can remove an integration-by-parts term.
- A separated solution provides a concrete check of an estimate.
- Energy decay describes a global quantity, not pointwise monotonicity.

**Review checklist**

- [x] I can state the equation and all data precisely.
- [x] I can recover the sine-mode calculation without notes.
- [?] I still need to justify the regularity assumptions.
- [ ] I can adapt the argument to a new boundary condition.

**Progress is supplied by the reader**

Use the bar to track a chosen page range, exercise list or reading plan. The values are entered manually; completing pages is distinct from understanding their proofs.

Reading progress 3/8

**LaTeX / Two editions from one source**

```
% Reading edition: hints and solutions are visible by default.
\documentclass{lumenbook}
% Self-test edition:
% \documentclass[solutions=false,hints=false]{lumenbook}
```